Weekly Progress Report

Name: K.Jeya Prawin Raj
Domain:Core Java development
Date of submission:14-03-2026

Week Ending: 01

I. Overview:

This week, the primary focus was on understanding the requirements of the Banking Information System project and setting up the development environment for building the prototype using Java. Initial efforts were made to study the project problem statement and design the basic architecture of the system. The core functionalities such as user registration, account management, deposit and withdrawal operations, fund transfer, and transaction history were analyzed.

II. Achievements:

1. Project Familiarization:

Carefully analyzed the Banking Information System problem statement and identified the key modules required for development.

Understood the workflow of a banking system including user registration, login authentication, account operations, and transaction tracking.

Planned the overall structure of the project using object-oriented programming concepts in Java.

2. Core Java Project Setup:

Name of the project: Banking Information System

Created the project folder and initialized the project using Java.

Implemented the basic project structure with the following classes:

- User.java
- Account.java
- Transaction.java
- BankingSystem.java

Implemented a basic console-based menu interface to simulate the banking system navigation.

3. Learning Core Java Concepts:

Revised important Core Java concepts required for the project such as:

i. Classes and Objects
ii. Constructors
iii. Java Collections (ArrayList)
iv. Exception Handling

Practiced simple Java programs to strengthen understanding of object-oriented programming concepts.

III. Challenges:

1. Designing the Application Structure:

Faced challenges in designing the relationships between classes such as User, Account, and Transaction.

Researched different approaches to organize the code structure effectively.

2. Data Storage Planning:

Explored methods to store account information and transaction data within the application.

Considered approaches such as ArrayList and file handling for temporary data storage during the prototype stage.

IV. Learning Resources:

1. Java Documentation:

Used official Java documentation to understand syntax and programming concepts.

Referred to online tutorials and guides for implementing object-oriented programming structures in Java.

2. Online Learning Platforms:

Watched programming tutorials related to Core Java and console-based applications.

V. Next Week's Goals:

1. User Registration Module:

Implement the user registration functionality.

Generate unique account numbers for users.

Store user details and account information in the system.

2. Account Operations:

Develop basic banking operations such as deposit and withdrawal.

Implement validation checks and error handling for invalid transactions.

VI. Additional Comments:

This week provided a strong foundation for the Banking Information System project. Understanding the problem statement and designing the project structure helped in planning the next stages of development. In the upcoming weeks, the focus will be on implementing the core banking functionalities and improving the usability of the system.